

Quantifying Atomically Dispersed Catalysts Using Deep Learning Assisted Microscopy

Haoyang Ni,[⊥] Zhenyao Wu,[⊥] Xinyi Wu,[⊥] Jacob G. Smith, Michael J. Zachman, Jian-min Zuo, Lili Ju, Guannan Zhang,* and Miaofang Chi*



Cite This: *Nano Lett.* 2023, 23, 7442–7448



Read Online

ACCESS |

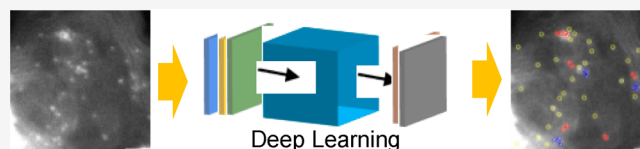
Metrics & More

Article Recommendations

Supporting Information

ABSTRACT: The catalytic performance of atomically dispersed catalysts (ADCs) is greatly influenced by their atomic configurations, such as atom–atom distances, clustering of atoms into dimers and trimers, and their distributions. Scanning transmission electron microscopy (STEM) is a powerful technique for imaging ADCs at the atomic scale; however, most STEM analyses of ADCs thus far have relied on human labeling, making it difficult to analyze large data sets. Here, we introduce a convolutional neural network (CNN)-based algorithm capable of quantifying the spatial arrangement of different adatom configurations. The algorithm was tested on different ADCs with varying support crystallinity and homogeneity. Results show that our algorithm can accurately identify atom positions and effectively analyze large data sets. This work provides a robust method to overcome a major bottleneck in STEM analysis for ADC catalyst research. We highlight the potential of this method to serve as an on-the-fly analysis tool for catalysts in future in situ microscopy experiments.

KEYWORDS: deep learning, STEM, image analysis, catalyst, convolutional neural network



Atomically dispersed catalysts (ADCs) have become a frontier in catalysis research since their first practical realization in 2011.^{1–5} Recent works have shown that the catalytic performance can be tuned by adjusting the number of atoms and their coordination configurations in few-atom clusters.^{6–8} For example, Pt–Ru dimers exhibit much higher hydrogen evolution activity and excellent stability compared to commercial Pt/C catalysts.⁹ Pt₂ and Fe₂ dimers on a mesoporous carbon nitride (mpg-C₃N₄) sample exhibit superior catalytic performance for the epoxidation of *trans*-stilbene to *trans*-stilbene oxide.^{7,10} However, the fabrication of thermally stable ADCs with designed adatom coordination is still challenging.¹¹ It is also difficult to uniformly manufacture these catalysts. Rather, it is common to find single atoms coexisting with dimers, trimers, and even much larger clusters in ADCs. The mixed configurations present a confounding variable when experimentally measuring the catalyst's activity and stability. Scanning transmission electron microscopy (STEM) characterization is often used to measure the distributions and concentrations of ADCs.^{12–14} The identification of various atomic configurations in STEM images has been performed through manual labeling or conventional computing methods.^{11,12,15,16} However, these methods are labor-intensive and limit the amount of data that can be processed. They are also prone to error due to human biases in data analysis, which highly depend on the experience of the microscope operator and their prior knowledge of microscopy and catalysts. An automatic method with minimal human intervention is necessary to efficiently analyze enough data to

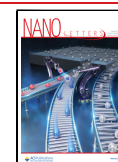
statistically quantify the adatom distribution. Such a method must be capable of accurately identifying individual adatoms decorating a substrate and extracting their locations and coordination with other adatoms.

Deep learning is a powerful technique for identifying features in images.^{17,18} Previous reports have demonstrated its potential for the analysis of atomic-resolution STEM images of crystalline materials.¹⁸ For example, convolutional neural networks (CNNs) were constructed and trained to identify and determine atomic column positions,¹⁹ classify different local structures,^{20–22} and quantify strain.²² However, these methods are not suitable for analyzing STEM images of ADCs on supports due to the different contrast schemes. Additionally, the commonly used STEM image simulation-based training method, which generates labeled images with well-defined atoms and defects,¹⁸ presents challenges for ADCs, because ADCs are anchored on substrates with varying and complex structures that are difficult to simulate. Recently, Mitchell et al.²³ reported a CNN-based algorithm for tracking and counting single-atom catalysts (SACs) that was trained using a set of 21 experimental images of Pt on nitrogen-doped carbon with human labeling. Their algorithm provides a

Received: May 21, 2023

Revised: August 2, 2023

Published: August 11, 2023



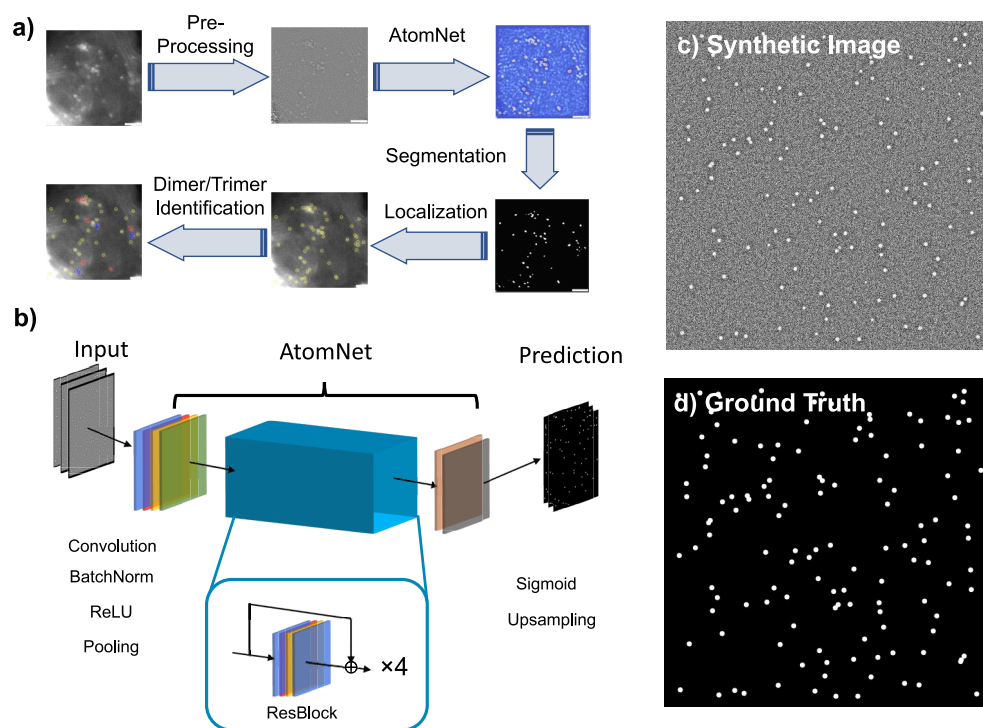


Figure 1. (a) Workflow of our atom recognition algorithm and (b) structure of AtomNet. Representative pair of synthetic image (c) and its ground truth (d) used for training.

valuable tool for single-atom analysis. In this work, we utilize a computer-generated training set consisting of 100000 synthetic images of ADCs and the corresponding ground truths. Compared to previous efforts, the utilization of a large training set allows us to thoroughly explore the parameter space without human bias, ensuring accuracy and generality. Further, the algorithm integrates multiple steps, including atom identification using an atom segmentation neural network, separation of overlapping atoms, analysis of the number of atoms in each ADC, and categorization of ADC configurations. This results in a fast-processing, CNN-based workflow that can identify and statistically quantify atomic configurations of ADCs, particularly the distribution of single atoms, dimers, and trimers, and enables on-the-fly analysis during in situ experiments.

■ BUILDING TRAINING DATA SETS AND WORKFLOW OF ATOM RECOGNITION ALGORITHM

Figure 1 summarizes the workflow of our algorithm. Raw STEM images I are initially preprocessed to remove the substrate contrast. The processed images E^* are then fed to AtomNet, a convolutional neural network built upon four consecutive residual blocks to extract deeper features (Figure 1b). AtomNet infers the probability of possible atom locations from processed STEM images and generates a soft prediction, S , and a probability threshold is applied to produce the binary image, H , that distinguishes atoms from background intensity and noise. The segmentation result then is sent to an atom localization module, wherein any overlapping atoms are separated and their locations are derived. The localization module requires prior knowledge of the typical atom size, which is estimated from the experimental conditions. Once all of the atom locations are obtained, the atom distribution

statistics are quantified to count the number of dimers and trimers based on interatomic distances.

Initially, we considered using manually labeled experimental STEM images²³ to train a CNN in our work. However, the small number of labeled images often limits the algorithm's generality and human labeling introduces bias. Nevertheless, we observed that by employing background subtraction, the slow-varying contrast from ADC substrates can be reduced. The resulting ADC image can be broadly described as adatoms superimposed on a high-frequency noise. Based on this observation, instead of simulating the ADC STEM images with detailed knowledge of ADC structure and composition or building a training set by human labeling, a synthetic data set consisting of images mimicking the background-subtracted ADC images was generated for training.

Our synthetic data set was produced in the following manner. First, a random number of coordinates was generated to represent the atom locations. To mimic the actual atoms in a preprocessed STEM image, each of these atoms was then blurred by a Gaussian kernel with a randomly chosen width. The blurring is done to model different STEM images acquired at different resolutions, magnifications, and defocusings. Since atoms in ADCs are mostly loaded on either the top or bottom surface of the sample, atoms observed in well-corrected in-focus STEM images could be assumed to be modeled as symmetric intensity blobs.

As ADF image signals are generated by converting electrons to electronic signals through a scintillator-photomultiplier, the noise level is mostly related to the electron dose and is best modeled by Poisson noise. We also include other types of noise, including Localvar, Speckle, and Salt&Pepper, that could be included due to model noise from electronic systems, such as dark current and random impulses. The ground truth images (GT) are generated by putting a circular mask with the same atomic position to generate the synthetic image. Once the

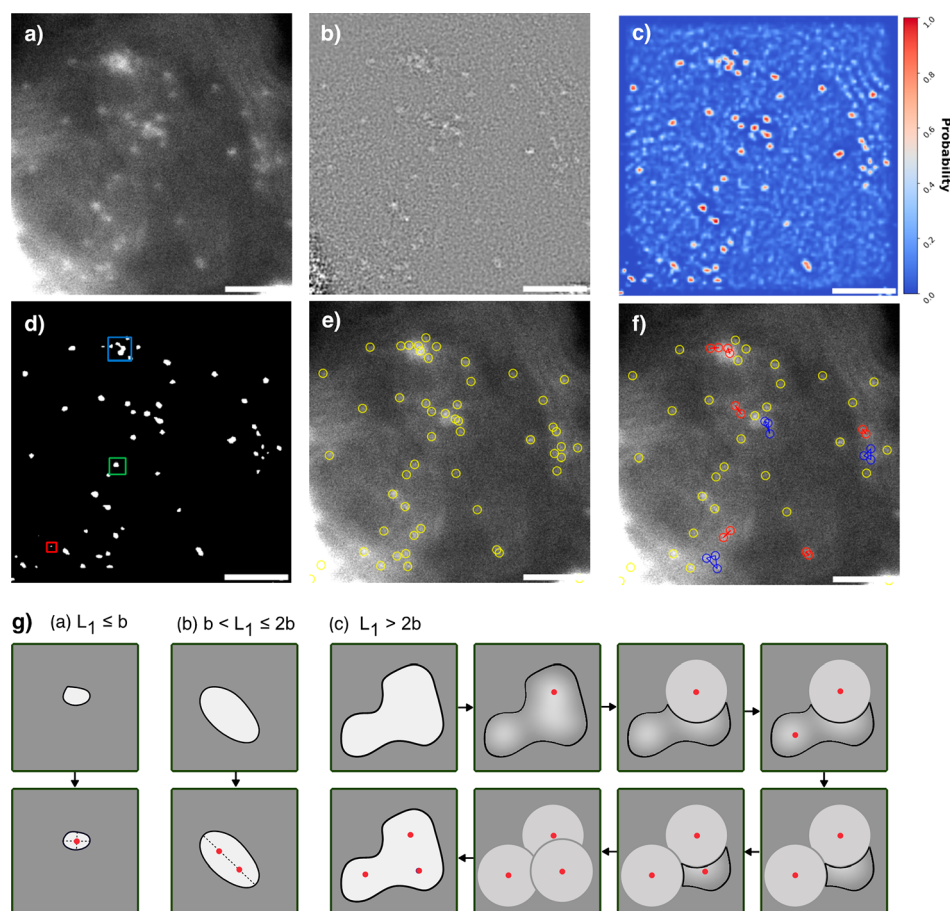


Figure 2. (a) Experimental STEM image of W/KB and (b) preprocessed image with subtracted background and normalization. (c) Inferred probability of atom in STEM image and (d) atoms segmented by threshold 0.5. Red, green, and blue boxes represent local noise, isolated atoms, and partially overlapping atoms, respectively. (e) Atoms localized using (g) illustrated procedure and (f) single atoms, dimers, and trimers identified and labeled. Scale bar, 2 nm.

synthetic data set was generated, AtomNet was trained by minimizing the difference between the soft prediction (S) and the GT using the mean squared error loss function.

TESTS ON EXPERIMENTAL STEM IMAGES

Our algorithm was first tested on experimental STEM images of W on a Ketjenblack carbon support (W/KB). Figure 2 illustrates the input and output of each step in our proposed algorithm. The raw STEM image (Figure 2a) is initially processed to remove the low-frequency contrast from the substrate by subtracting the raw image from a Gaussian-filtered image (Figure 2b) created with a Sigma of 10 pixels, which is more than 3 times the size of an atom radius (<3 pixels in radius). The image is then normalized to a range of 0 to 1.

AtomNet infers the probability of an atom within the processed image to generate a probability map (Figure 2c). Despite the blurring and noise seen in Figure 2b, the probability map (Figure 2c) shows localized and highlighted features that correspond to adatoms. We chose a probability threshold of 0.5 to binarize the probability map and segment the atoms from background features. The segmentation image may contain three possible features, labeled in the red, green, and blue boxes in Figure 2d. Isolated atoms appear as circular or elliptical regions (green box), while regions with only a few pixels (red box) and large blobs with irregular shapes (blue box) also exist.

To handle these possible features and extract the 2D atom coordinates, an atom localization procedure is proposed in Figure 2g. First, an area threshold is applied to the binarized image (Figure 2d) to eliminate false positives caused by noise. Then, for regions that are accepted, the shape of the atom is evaluated by calculating the major axis length L_1 and comparing it with a length threshold b , which could be estimated from atom size. For $L_1 \leq b$, we consider the region to be a single atom, and the coordinate of the atom is the peak position of this region in the probability map. For $b < L_1 \leq 2b$, the region is a group of two atoms, where their locations are at the two trisection points of the major axis. For $L_1 > 2b$, the region has three or more atoms, and the location of each atom is determined using soft prediction S . We first locate the point with the highest probability in S as the center of an atom with radius b . This region will be masked and get updated. Then the highest probability in the updated region is used as the atom position.

Therefore, the key to successfully extracting of atom locations from the segmentation image is to determine the size threshold and the length threshold b , both of which can be derived from the experimental conditions. In STEM images, apparent atom size is primarily determined by the electron probe size and the real space sampling. Consequently, regions too small to correspond to an atom are rejected as noise. Since b is used to differentiate single atoms and overlapping

neighboring atoms, a good estimation of b is three times the atom radius, considering a 50% overlap. With b determined, Figure 2e shows an identification result, with each atom marked by yellow circles. Comparing Figure 2a and Figure 2e, we see that most atoms easily recognized by humans are detected. A good segmentation result shows that choosing parameters for atom localization based on prior knowledge of the experimental conditions is useful. Monomers, dimers, and trimers can be detected by calculating atomic pair distances. While a clearly defined critical spacing between W – W atoms for dimers and trimers does not exist, we use 5 Å as the critical spacing to categorize dimers or trimers for demonstration of the algorithm. A different spacing value will be used for analyzing a different ADC system later in this work. Figure 2f shows the analysis results of W /KB. Monomers, dimers, and trimers are labeled by yellow, red, and blue circles, respectively.

To further explore the generality of our proposed algorithms, experimental images from Pt on graphitic carbon nitride (Pt/CN) and Ir on MgO (Ir/MgO) were analyzed by using the same approach. The accuracy of our algorithm is quantified by comparing the atom identification result against human labeling. Figure 3 shows the result of human labeling of

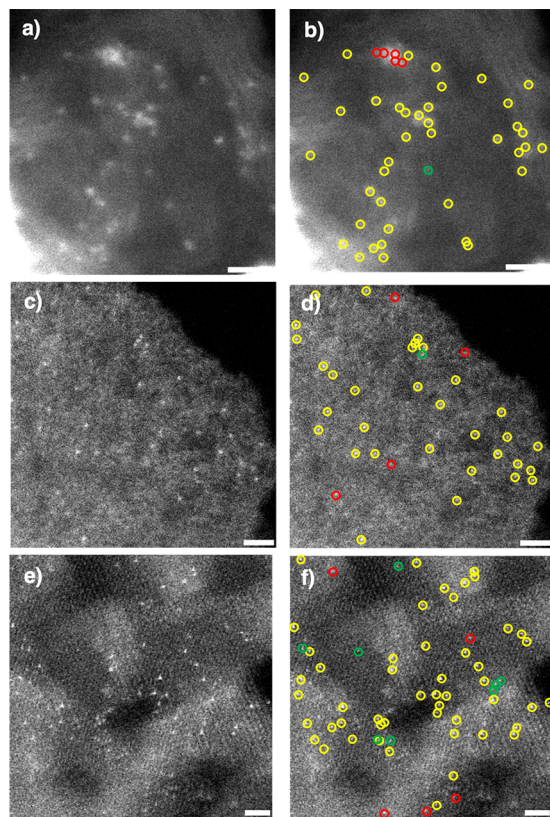


Figure 3. Raw STEM images of W /KB (a), Pt/CN (c), and Ir/MgO (e) and corresponding human labeling (b, d, f). Red, yellow, and green circles label the false positive, true positive, and false negative atoms identified by computer algorithm. Scale bar, 2 nm.

STEM images of ADCs. In Figure 3d–f, atoms recognized by both algorithm and human (measured true positive, TP) are labeled by yellow circles, with atoms identified by our algorithm but not by a human (false positive, FP) as red circles and atoms missed by the algorithm (false negative, FN) as green circles.

To quantify the accuracy of our algorithm, metrics of precision ($TP/(TP + FP)$) and recall ($TP/(TP + FN)$) can be calculated. Precision quantifies the tendency for an algorithm to reject unreal atoms, and recall quantifies how correctly the algorithm identifies the real atoms. Table 1 summarizes the

Table 1. Precision and Recall of Our Algorithm for W /KB, Pt/CN, and Ir/MgO

Image	Precision	Recall
W /KB	0.886	0.975
Pt/CN	0.889	0.970
Ir/MgO	0.911	0.864

precision and recall of each image. We show that in all three images a precision and recall of approximately 0.9 can be achieved, indicating that our algorithm can detect atoms with good confidence compared to human vision. These results also show that our algorithm can be applied to STEM images of different catalyst configurations and taken under different imaging conditions without any retraining of our network. As a result, we expect our algorithm to be generally applicable to ADCs STEM image analysis. We also note that, for a real sample, there is no guarantee that human interpretation is correct and that there might be some disagreement between different operators.

It should be noted that our method is a contrast-based atom recognition method designed for high-angle annular dark field (HAADF) STEM images. Therefore, a reasonable visibility of adatoms in ADCs is necessary. In HAADF images, image intensities scale with atomic number Z raised to a power of 1.6–2.0. Our method is thus effective when adatoms in ADCs are relatively heavier than substrates. When adatoms have similar or smaller atomic numbers than substrates, such as Cu on CeO_2 ,²⁴ or when the support is much thicker and the effective projected atomic number of the support is larger than that of the metal atoms, identifying adatoms becomes challenging. Additionally, adatoms embedded in a thick matrix or those out-of-focus can contribute a weaker and ambiguous contrast, leading to low certainty in atom identification. Training neural networks with data sets that include substrate contrast and electron optics imperfections could enhance the accuracy and generality of the current algorithm.

■ STATISTICAL ANALYSIS OF STEM IMAGES

Electron microscopy for catalyst analysis often faces criticism for its limited sampling. Despite its power, the technique's drawback lies in its restricted sampling capability. Additionally, analyzing multiple images becomes time-consuming. To demonstrate our algorithm's ability to overcome these challenges, we conducted a case study on a Pt/CN catalyst using 11 ADF STEM images from various CN flakes. In the 11 images, a total of 531 Pt atoms were identified and analyzed (Figure 4a and images in the Supporting Information). To explore the distribution of interatomic distances, we calculated a histogram of the interatomic distances between Pt–Pt atoms (Figure 4b). We observe a major peak at 5 Å. A detailed distribution was drawn from the 0 to 2 nm range, as shown in Figure 4b inset, which shows a nonzero peak at ~ 1 Å with fluctuations around 5 Å to 1 nm, indicating that different regions have different atom distributions. The distribution of Pt–Pt distances indicates that on average Pt atoms are separated 5 Å apart. While the definition of the metal–metal

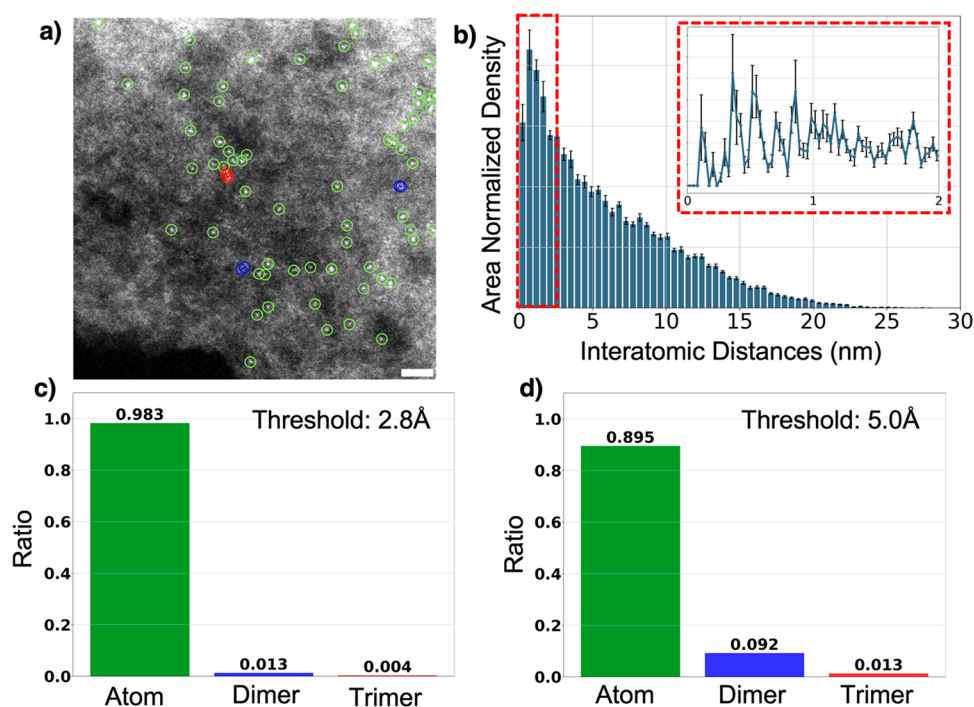


Figure 4. (a) Recognition of single atoms (green circles), dimers (blue circles), and trimers (red circles) in a representative Pt/CN STEM image. Scale bar, 2 nm. (b) Area normalized density of Pt–Pt pair distribution. Inset shows a zoomed-in distribution of atoms within 0–2 nm. (c, d) Ratio of Pt atoms forming single atoms, dimers, and trimers in selected 11 STEM images, by using Pt–Pt bond lengths of 2.8 Å (c) and 5.0 Å (d) as a threshold.

bonding length in dimers and trimers is not present, the flexibility of taking user input is necessary. Using 2.8 Å (Pt–Pt bond length in single-crystal Pt metal) as a threshold, we identify the possible dimers and trimers in the ADF images. For example, in Figure 4a, three Pt atoms forming a trimer and two pairs of dimers are detected. Among the 531 Pt atoms identified, 98.3% of atoms are monomers with 1.3% forming dimers and 0.4% forming trimers (Figure 4c). However, it is known that the bond length between individual metal atoms in ADCs is larger than in their bulk counterparts and varies with various parameters, such as reaction environments, support oxide, and metal species. Considering that there is currently no well-defined definition of bonding length for ADCs, it is critical that the algorithm provides flexibility for the user to define the bonding length and generate quantitative statistics accordingly. As shown in Figure 4d, by choosing 5.0 Å as a threshold, the ratio of single atoms reduces to 89.5%, and the ratio of forming dimers and trimers increases to 9.2% and 1.3%. The statistics here indicate that this specific Pt/CN sample contains a majority of single-atom configurations, with less than 5% of the observed adatom configurations consisting of dimers, trimers, or clusters.

It should be highlighted that the analysis of 11 images with a size of 1024 pixel² using GPU takes a total of 4 seconds, averaging around 0.36 s/image. In comparison, acquiring a regular STEM-HAADF image typically takes around 6 s with a size of 1024 pixel² and a 6 μ s dwell time to ensure an adequate signal-to-noise ratio. Even with a fast dwell time of 1 μ s, it still takes 1 s to acquire each image, which is longer than the algorithm's processing time. This indicates that the algorithm can be seamlessly integrated on-the-fly into in situ STEM experiments to study the dynamic evolution of catalysts under realistic reaction conditions. Such studies are crucial for identifying the active sites and intermediate states.

In summary, we have developed an innovative machine-learning-assisted algorithm for quantifying atom configurations, extracting atom coordinates, and generating atom distribution statistics (e.g., single atoms, dimers, or trimers) in atomically dispersed catalysts (ADCs). Our algorithm demonstrates excellent performance in identifying atoms from background contrast and noise in preprocessed STEM images. We have thoroughly validated the robustness of our method across various configurations, imaging conditions, and materials systems, including catalysts on supports with inhomogeneous thickness and lattice contrast. Notably, our algorithm enables efficient and rapid analysis of large image data sets without requiring network retraining. This approach holds the potential to significantly enhance the efficiency and accuracy of ADC analysis, alleviate the sampling challenges in TEM, and facilitate on-the-fly analysis for in situ electron microscopy of ADCs.

METHODS

STEM Imaging. STEM specimens were prepared by dry casting catalyst powders on lacey carbon grids. The exposure time to ambient conditions was minimized in order to reduce the potential contamination of the specimens. W/KB and Pt/CN imaging were performed on a JEOL JEM-ARM200F (NeoARM) scanning transmission electron microscope (STEM) at ORNL's Center of Nanophase Materials Sciences, equipped with an integrated ASCOR (Cs) probe corrector. Both W/KB and Pt/CN STEM images were taken at 80 kV with a 40 μ m aperture (~ 27 mrad) and 8 cm camera length. W/KB images are acquired with a total dose of 1.6×10^5 e/ $\text{\AA}^2$ and Pt/CN images with a total dose of 1.6×10^4 e/ $\text{\AA}^2$. The Ir/MgO Image was acquired at 200 kV on an FEI Titan 80/300S microscope equipped with a probe corrector with a

convergence angle of 30 mrad and a total dose of 3.0×10^4 e/ $\text{\AA}^2$.

AtomNet. The architecture of AtomNet can be seen in Figure 1b. An input grayscale image was first fed into a 7×7 convolution layer, then a batch normalized (BatchNorm) layer, and a rectified linear unit activation (ReLU) layer. After pooling, the output goes through four consecutive ResBlock units containing two 3×3 convolution layers at the beginning and the end of ResBlock, with BatchNorm and ReLU in the middle. A residual connection is utilized at the end of each ResBlock. The extracted features from four ResBlock units were then activated by a Sigmoid layer to produce the soft probability map. Finally the probability map is up-sampled to the size of the input image.

Training Strategy. The training set contains 100000 synthetic images with a randomized number of atoms from 40 to 200, Gaussian blurring with variance σ from 1 to 8, and intensity from 0 to 100. Then Poisson noise, Localvar noise, Speckle noise and Salt&Pepper noise were added to the synthetic images.

AtomNet was trained on our synthetic data set for 2500 iterations optimized using stochastic gradient descent (SGD) with a momentum of 0.9. The batch size is 16. The initial learning rate is 0.0001 and decreases following poly learning rate policy²⁵ with a power of 0.9. Mean squared error loss function is used for minimization of the distance between prediction S and ground truth GT . To quantitatively evaluate the trained model, we also create a validation set which consists of 200 synthetic images.

■ ASSOCIATED CONTENT

Data Availability Statement

The algorithm can be downloaded at the following link: <https://github.com/ornlstemgroup/ORN-ADC-Deep-Learning>.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.3c01892>.

Plot of training loss and validation accuracy curve for AtomNet; 11 Pt/CN images used for statistical analysis with identified atoms (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Guannan Zhang — Computer Science and Mathematics Division, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; Email: zhangg@ornl.gov

Miaofang Chi — Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0003-0764-1567; Email: chim@ornl.gov

Authors

Haoyang Ni — Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; Department of Materials Science and Engineering, University of Illinois at Urbana–Champaign, Urbana, Illinois 61801, United States

Zhenyao Wu — Department of Mathematics, University of South Carolina, Columbia, South Carolina 29208, United States

Xinyi Wu — Department of Mathematics, University of South Carolina, Columbia, South Carolina 29208, United States

Jacob G. Smith — Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States

Michael J. Zachman — Center for Nanophase Materials Sciences, Oak Ridge National Laboratory, Oak Ridge, Tennessee 37831, United States; orcid.org/0000-0003-1910-1357

Jian-min Zuo — Department of Materials Science and Engineering, University of Illinois at Urbana–Champaign, Urbana, Illinois 61801, United States; orcid.org/0000-0002-5151-3370

Lili Ju — Department of Mathematics, University of South Carolina, Columbia, South Carolina 29208, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.nanolett.3c01892>

Author Contributions

[†]H.N., Z.W., and X.W. contributed equally to this work.

Funding

This manuscript has been authored by UT-Battelle, LLC under Contract DE-AC05-00OR22725 with the U.S. Department of Energy. The United States Government retains and the publisher, by accepting the article for publication, acknowledges that the United States Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this manuscript, or allow others to do so, for United States Government purposes. The Department of Energy will provide public access to these results of federally sponsored research in accordance with the DOE Public Access Plan (<http://energy.gov/downloads/doe-public-access-plan>).

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was conducted as part of an Early Career Award supported by the U.S. Department of Energy, Office of Basic Energy Sciences (DOE-BES), and by the DOE-BES, Division of Materials Sciences and Engineering. Algorithm developments were supported by the DOE Advanced Scientific Computing Research, Applied Mathematics Program. Microscopy experiments were performed at the Center for Nanophase Materials Sciences, a U.S. DOE Office of Science User Facility at Oak Ridge National Laboratory.

■ REFERENCES

- (1) Qiao, B.; Wang, A.; Yang, X.; Allard, L. F.; Jiang, Z.; Cui, Y.; Liu, J.; Li, J.; Zhang, T. Single-Atom Catalysis of CO Oxidation Using Pt1/FeOx. *Nat. Chem.* **2011**, 3 (8), 634–641.
- (2) Gates, B. C.; Flytzani-Stephanopoulos, M.; Dixon, D. A.; Katz, A. Atomically Dispersed Supported Metal Catalysts: Perspectives and Suggestions for Future Research. *Catal. Sci. Technol.* **2017**, 7, 4259–4275.
- (3) Zhang, H.; Liu, G.; Shi, L.; Ye, J. Single-Atom Catalysts: Emerging Multifunctional Materials in Heterogeneous Catalysis. *Adv. Energy Mater.* **2018**, 8, 1701343.
- (4) Cheng, N.; Zhang, L.; Doyle-Davis, K.; Sun, X. Single-Atom Catalysts: From Design to Application. *Electrochem. Energy Rev.* **2019**, 2, 539–573.

- (5) Yang, X. F.; Wang, A.; Qiao, B.; Li, J.; Liu, J.; Zhang, T. Single-Atom Catalysts: A New Frontier in Heterogeneous Catalysis. *Acc. Chem. Res.* **2013**, *46* (8), 1740–1748.
- (6) An, S.; Zhang, G.; Wang, T.; Zhang, W.; Li, K.; Song, C.; Miller, J. T.; Miao, S.; Wang, J.; Guo, X. High-Density Ultra-Small Clusters and Single-Atom Fe Sites Embedded in Graphitic Carbon Nitride (g-C₃N₄) for Highly Efficient Catalytic Advanced Oxidation Processes. *ACS Nano* **2018**, *12* (9), 9441–9450.
- (7) Tian, S.; Fu, Q.; Chen, W.; Feng, Q.; Chen, Z.; Zhang, J.; Cheong, W. C.; Yu, R.; Gu, L.; Dong, J.; Luo, J.; Chen, C.; Peng, Q.; Draxl, C.; Wang, D.; Li, Y. Carbon Nitride Supported Fe₂ Cluster Catalysts with Superior Performance for Alkene Epoxidation. *Nature Communications* **2018** *9*:1 **2018**, *9* (1), 1–7.
- (8) Zhang, L.; Si, R.; Liu, H.; Chen, N.; Wang, Q.; Adair, K.; Wang, Z.; Chen, J.; Song, Z.; Li, J.; Banis, M. N.; Li, R.; Sham, T. K.; Gu, M.; Liu, L. M.; Botton, G. A.; Sun, X. Atomic Layer Deposited Pt-Ru Dual-Metal Dimers and Identifying Their Active Sites for Hydrogen Evolution Reaction. *Nature Communications* **2019** *10*:1 **2019**, *10* (1), 1–11.
- (9) Yang, X.; Adair, K. R.; Gao, X.; Sun, X. Recent Advances and Perspectives on Thin Electrolytes for High-Energy-Density Solid-State Lithium Batteries. *Energy Environ. Sci.* **2021**, *14* (2), 643–671.
- (10) Tian, S.; Wang, B.; Gong, W.; He, Z.; Xu, Q.; Chen, W.; Zhang, Q.; Zhu, Y.; Yang, J.; Fu, Q.; Chen, C.; Bu, Y.; Gu, L.; Sun, X.; Zhao, H.; Wang, D.; Li, Y. Dual-Atom Pt Heterogeneous Catalyst with Excellent Catalytic Performances for the Selective Hydrogenation and Epoxidation. *Nat. Commun.* **2021**, *12* (1), 3181.
- (11) Cai, X.; Chen, X.; Ying, Z.; Wang, S.; Chen, Y.; Cai, Y.; Long, G.; Liu, H.; Wang, N. Towards a Library of Atomically Dispersed Catalysts. *Mater. Des.* **2021**, *210*, 110080.
- (12) Tieu, P.; Yan, X.; Xu, M.; Christopher, P.; Pan, X. Directly Probing the Local Coordination, Charge State, and Stability of Single Atom Catalysts by Advanced Electron Microscopy: A Review. *Small* **2021**, *17*, 2006482.
- (13) Gao, W.; Hood, Z. D.; Chi, M. Interfaces in Heterogeneous Catalysts: Advancing Mechanistic Understanding through Atomic-Scale Measurements. *Acc. Chem. Res.* **2017**, *50* (4), 787–795.
- (14) Chao, H.-Y.; Venkatraman, K.; Moniri, S.; Jiang, Y.; Tang, X.; Dai, S.; Gao, W.; Miao, J.; Chi, M. *In Situ* and Emerging Transmission Electron Microscopy for Catalysis Research. *Chem. Rev.* **2023**, *123*, 8347.
- (15) Liu, S.; Xu, H.; Liu, D.; Yu, H.; Zhang, F.; Zhang, P.; Zhang, R.; Liu, W. Identify the Activity Origin of Pt Single-Atom Catalyst via Atom-by-Atom Counting. *J. Am. Chem. Soc.* **2021**, *143* (37), 15243–15249.
- (16) Li, Y.; Shan, W.; Zachman, M. J.; Wang, M.; Hwang, S.; Tabassum, H.; Yang, J.; Yang, X.; Karakalos, S.; Feng, Z.; Wang, G.; Wu, G. Atomically Dispersed Dual-Metal Site Catalysts for Enhanced CO₂ Reduction: Mechanistic Insight into Active Site Structures. *Angew. Chem., Int. Ed.* **2022**, *61* (28), e202205632.
- (17) Spurgeon, S. R.; Ophus, C.; Jones, L.; Petford-Long, A.; Kalinin, S. V.; Olszta, M. J.; Dunin-Borkowski, R. E.; Salmon, N.; Hattar, K.; Yang, W.-C. D.; Sharma, R.; Du, Y.; Chiaramonti, A.; Zheng, H.; Buck, E. C.; Kovarik, L.; Penn, R. L.; Li, D.; Zhang, X.; Murayama, M.; Taheri, M. L. Towards Data-Driven next-Generation Transmission Electron Microscopy. *Nat. Mater.* **2021**, *20* (3), 274–279.
- (18) Kalinin, S. V.; Ophus, C.; Voyles, P. M.; Erni, R.; Kepaptsoglou, D.; Grillo, V.; Lupini, A. R.; Oxley, M. P.; Schwenker, E.; Chan, M. K. Y.; Etheridge, J.; Li, X.; Han, G. G. D.; Ziatdinov, M.; Shibata, N.; Pennycook, S. J. Machine Learning in Scanning Transmission Electron Microscopy. *Nature Reviews Methods Primers* **2022**, *2* (1), 11.
- (19) Lin, R.; Zhang, R.; Wang, C.; Yang, X.-Q.; Xin, H. L. TEMImageNet Training Library and AtomSegNet Deep-Learning Models for High-Precision Atom Segmentation, Localization, Denoising, and Deblurring of Atomic-Resolution Images. *Sci. Rep.* **2021**, *11* (1), 5386.
- (20) Maxim, Z.; Jesse, S.; Sumpter, B. G.; Kalinin, S. V.; Dyck, O. Tracking Atomic Structure Evolution during Directed Electron Beam Induced Si-Atom Motion in Graphene via Deep Machine Learning. *Nanotechnology* **2021**, *32* (3), 035703.
- (21) Ziatdinov, M.; Dyck, O.; Maksov, A.; Li, X.; Sang, X.; Xiao, K.; Unocic, R. R.; Vasudevan, R.; Jesse, S.; Kalinin, S. V. Deep Learning of Atomically Resolved Scanning Transmission Electron Microscopy Images: Chemical Identification and Tracking Local Transformations. *ACS Nano* **2017**, *11* (12), 12742–12752.
- (22) Lee, C.-H.; Khan, A.; Luo, D.; Santos, T. P.; Shi, C.; Janicek, B. E.; Kang, S.; Zhu, W.; Sobh, N. A.; Schleife, A.; Clark, B. K.; Huang, P. Y. Deep Learning Enabled Strain Mapping of Single-Atom Defects in Two-Dimensional Transition Metal Dichalcogenides with Sub-Picometer Precision. *Nano Lett.* **2020**, *20* (5), 3369–3377.
- (23) Mitchell, S.; Parés, F.; Faust Akl, D.; Collins, S. M.; Kepaptsoglou, D. M.; Ramasse, Q. M.; Garcia-Gasulla, D.; Pérez-Ramírez, J.; López, N. Automated Image Analysis for Single-Atom Detection in Catalytic Materials by Transmission Electron Microscopy. *J. Am. Chem. Soc.* **2022**, *144* (18), 8018–8029.
- (24) Sun, Y.; Polo-Garzon, F.; Bao, Z.; Moon, J.; Huang, Z.; Chen, H.; Chen, Z.; Yang, Z.; Chi, M.; Wu, Z.; Liu, J.; Dai, S. Manipulating Copper Dispersion on Ceria for Enhanced Catalysis: A Nanocrystal-Based Atom-Trapping Strategy. *Advanced Science* **2022**, *9* (8), 2104749.
- (25) Zhao, H.; Shi, J.; Qi, X.; Wang, X.; Jia, J. Pyramid Scene Parsing Network. In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*; IEEE, 2017; pp 6230–6239. DOI: 10.1109/CVPR.2017.660.

Recommended by ACS

Harnessing Neural Networks for Elucidating X-ray Absorption Structure–Spectrum Relationships in Amorphous Carbon

Hyuna Kwon, Tuan Anh Pham, *et al.*

AUGUST 15, 2023

THE JOURNAL OF PHYSICAL CHEMISTRY C

READ 

Machine Learning Approach to Enable Spectral Imaging Analysis for Particularly Complex Nanomaterial Systems

Haili Jia, Paulette Clancy, *et al.*

DECEMBER 20, 2022

ACS NANO

READ 

Machine Learning Surrogate Model for Acceleration of Ferroelectric Phase-Field Modeling

Kévin Alhada-Lahbabi, Brice Gautier, *et al.*

JULY 10, 2023

ACS APPLIED ELECTRONIC MATERIALS

READ 

Automated Experiment in 4D-STEM: Exploring Emergent Physics and Structural Behaviors

Kevin M. Roccapiore, Sergei V. Kalinin, *et al.*

APRIL 27, 2022

ACS NANO

READ 

Get More Suggestions >